Lab 3 **Statistics for Laboratory Scientists II**
Solutions

1. Preparations:

```
# read in the data
lab3 <- read.csv("lab3.csv")

# plot the data
plot(response ~ concentration, data=lab3)
```

There's an obvious curvature; if you look closely at the lab, you can see that I suggest taking $\log_{10}$ of the response.

```
# add a log10(response) column
lab3$y <- log10(lab3$response)

# now plot the data
plot(y ~ concentration, data=lab3)
```

You can see that the curvature is now gone.

(a) Fit the model and get estimates of β_0, β_1, σ.

```
# Fit the linear model
lm.out <- lm(y ~ concentration, data=lab3)

# Get the summary
lm.sum <- summary(lm.out)
lm.sum
```

$\hat{\beta}_0$ = 1.996 (SE = 0.009), $\hat{\beta}_1$ = –0.0204 (SE = 0.0003), $\hat{\sigma}$ = 0.022

(b) Get 95% confidence intervals for β_0 and β_1.

```
# degrees of freedom
df <- lm.out$df  # (22)
tm <- qt(0.975, df) # quantile of t distribution

# estimates and SEs
b0 <- lm.sum$coef[1,1]
se0 <- lm.sum$coef[1,2]
b1 <- lm.sum$coef[2,1]
se1 <- lm.sum$coef[2,2]

# the 95% CIs
b0 + tm * c(-1,1) * se0
b1 + tm * c(-1,1) * se1
```

95% CI for β_0 = (1.98, 2.02); 95% CI for β_1 = (–0.0211, –0.0197)

(c) Assess the appropriateness of the model: look at the residuals versus the fitted values and a QQ-plot of the residuals.

```
# the easy way
plot(lm.out)
```

```
# the direct way
plot(fitted(lm.out), resid(lm.out))

qqnorm(resid(lm.out))
qqline(resid(lm.out))
```

This looks reasonable—we see little evidence against constant variance or against normality of errors.

2. Now onto the calibration. First, get the calibrate function in the file `calibrate.R`; download the file and then "source" it.

```
# load the calibrate() function
source("calibrate.R")
```

Now do everything else in one step; make sure you do this in the $\log_{10}$ scale.

```
calibrate(lab3$concentration, lab3$y, log10(35.6))
```

The answer: the estimate is 21.8 ppm, with a 95% confidence interval of (19.5, 24.2).